

Shaik Mohammad Fardeen

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SOFTWARE DEVELOPER

Education

George Mason University, College of Engineering

Aug 2025 – May 2027

M.S. Computer Science

Coursework: Intro Artificial Intelligence, Software Eng for WWW, Component based Software Devolepment, Analysis of Algorithms, Computer Systems and System Programming, Mathematical Foundations of CS, Database Ssytems, Mobile Immersive Computing, Modern Computer Architecture

Vellore Institute of Technology

Sept 2021 – May 2025

B.Tech. Computer Science and Engineering

Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Systems, Software Engineering, AI & ML

Technical Skills

Languages: Python, JavaScript, Java, SQL, R, HTML, CSS

Tools: TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy, Matplotlib, Jupyter Notebooks

Core Areas: Machine Learning, Deep Learning, Neural Networks, NLP, Computer Vision, Statistics, Data Analysis, Feature Engineering

Professional Experience

Quadrant Technologies | Cloud AI Intern | Redmond, WA

Jun 2026 – Aug 2026

- Developed an explainable candidate ranking system using semantic search and LLMs with adjustable scoring weights across resume; implemented blind review mode showing exactly where each matched skill, ensuring full transparency trust in AI decisions.
- Built Screening Assistant using React, FastAPI, and Azure services to automate resume parsing; leveraged Azure AI Document Intelligence and Azure OpenAI for intelligent extraction enabling recruiters to reduce manual screening time significantly.

Ethnus Codemithra | Full-Stack Developer Intern

Aug 2023 – Nov 2023

- Enhanced UI/UX by refactoring reusable React components and optimizing state management, resulting in 15% increased user engagement; implemented automated tests with Jest/React Testing Library, API monitoring with Postman Collections, and achieved 80

Projects

AI Contract Invoice Intelligence System | Apache Airflow, FastAPI APIs, PDF Processing Pipeline, Database Integration **Nov 2025**

- Built scalable NLP pipelines using Python and TensorFlow to automate invoice extraction and data enrichment from unstructured PDFs; successfully processed 150,000+ invoices annually with 98% accuracy, verified through quarterly QA audits.
- Engineered Apache Airflow DAG-based orchestration workflow covering end-to-end ingestion, text extraction, validation, and storage pipelines; implemented automated validation rules and error detection that reduced data quality issues by 60%.

Blood Bank Management System | React, FastAPI, MySQL, Google Cloud Platform (GCP), RESTful APIs

Oct 2025

- Developed full-stack cloud-based donor registration and blood inventory management system using React, FastAPI, and MySQL, deployed on Google Cloud Platform with real-time availability search and request tracking.
- Engineered RESTful APIs for donor onboarding, blood donation history, and inventory management; implemented Google Cloud Storage integration for document handling and optimized database queries reducing query latency by 40%.

Loan Default Prediction | Streamlit, Python Backend, SHAP/LIME Explainability, Data Visualization, PyTorch, XGBoost **Feb 2026**

- Developed an ensemble machine learning model combining PyTorch neural networks and XGBoost gradient boosting to predict loan defaults; performed extensive feature engineering, hyperparameter tuning, and cross-validation to achieve high predictive accuracy.
- Deployed an interactive Streamlit web application powered by SHAP and LIME explainability frameworks, enabling non-technical banking professionals to understand model predictions and make transparent, data-driven lending decisions with full audit trails.

Autism Detection | Flask Web Framework, CNN/LSTM Models, MySQL Database, Frontend UI, Python, TensorFlow

Apr 2026

- Designed and trained CNN/LSTM hybrid deep learning architecture for autism spectrum disorder screening; enhanced baseline model accuracy by 15% to achieve 91% using advanced training techniques learning-rate scheduling, and class-weight balancing.
- Built end-to-end deep learning pipeline with data preprocessing, model training, validation, and evaluation on 1,500+ labeled medical samples; integrated Flask backend with MySQL database for patient data management and integrated model inference.

Research

Hybrid Algorithm for Parkinson's Prediction | Python, Scikit-Learn (Random Forest, SVM), SQL.

Nov 2025

- Presented hybrid machine learning model at Springer 2024 combining Random Forest and SVM algorithms, achieving 88.7% accuracy on UCI Parkinson's dataset (195 instances) with 6-8% improvement over individual models by optimizing sensitivity and specificity for early-stage neurological disorder detection.

Leadership

IEEE Computer Society — Events Head (Mar 2022 – Dec 2024): Led large-scale technical events and hackathons with participation exceeding 700 students.